

FRACMATH: numerical damage update

The diagram shows the default modified von Mises driver. The CPU Rankine-stress example replaces this calculation while keeping the other operations. The modified von Mises driver used below is

$$\mathcal{E}_{\text{mvM}}(I_1, J_2) = \frac{k-1}{2k(1-2\nu)} I_1 + \frac{1}{2k} \sqrt{\left(\frac{k-1}{1-2\nu}\right)^2 I_1^2 + \frac{12k}{(1+\nu)^2} J_2}, \quad k = f_c/f_t.$$

E is Young's modulus; ν is Poisson's ratio; f_t and f_c are tensile and compressive strengths; G_F is fracture energy per crack area. Engineering shear is γ_{xy} and tensor shear is $\gamma_{xy}/2$. The diagram shows the Oliver-width implementation. m is the load increment and $\mathbf{n}$ is the crack-normal unit vector. $\mathbf{D}_0$ is the plane-stress matrix form of the elastic tensor $\mathbb{C}_0$, acting on $[\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy}]^T$.

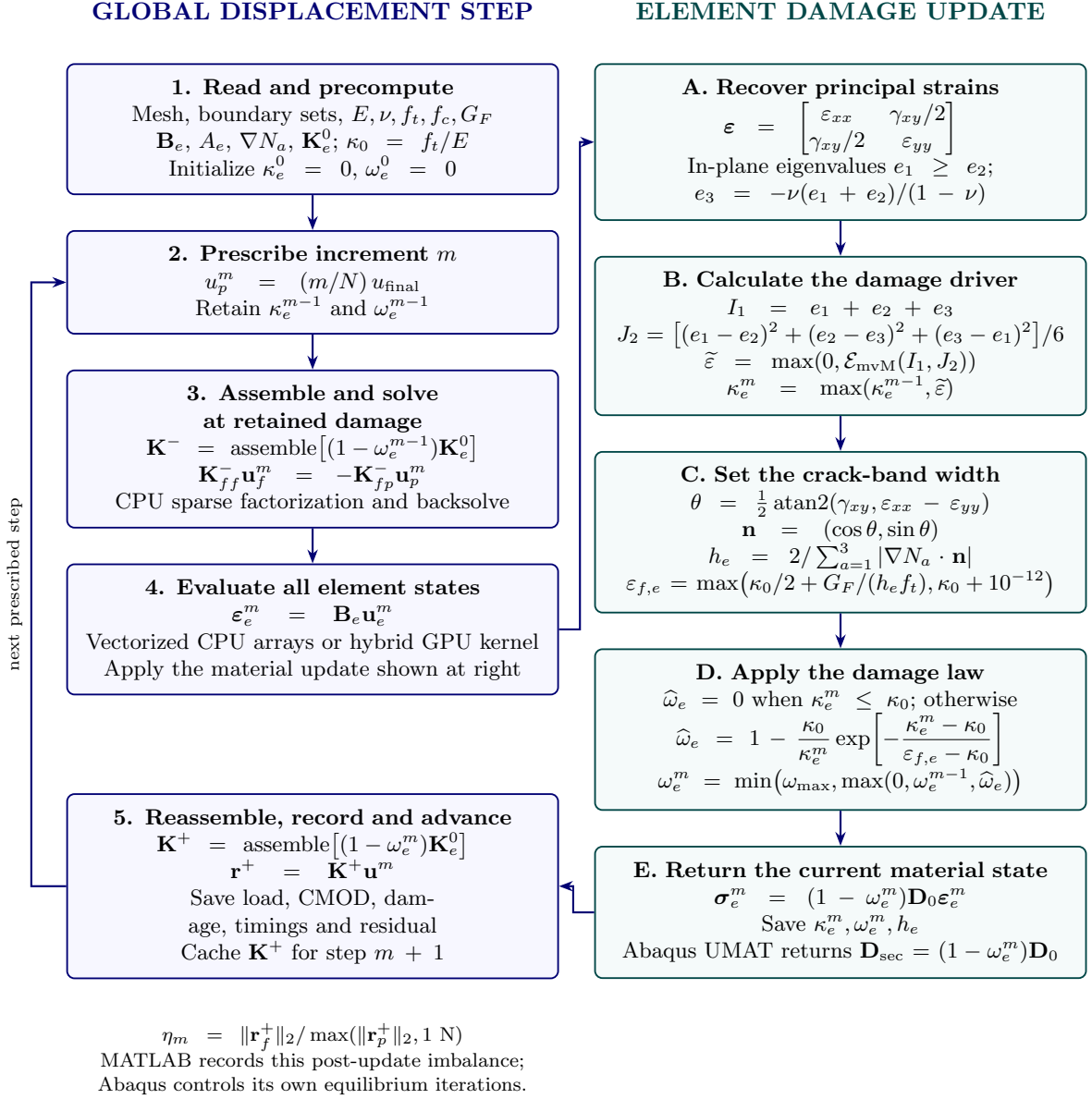


Figure 1: Step-by-step 2D damage calculation and sequential MATLAB implementation, drawn directly in LaTeX. Subscripts e, f, p denote element, free and prescribed degrees of freedom; m is the increment index and N its prescribed total. $\mathbf{D}_0$ and $\mathbf{K}_e^0$ are elastic material and element matrices; $\mathbf{B}_e$ maps element displacements to strain. $\hat{\omega}_e$ is the trial damage and $\omega_{\text{max}} = 1 - 10^{-12}$ is the 2D damage cap. The UMAT uses the same material equations but Abaqus supplies trial strains and controls nonlinear equilibrium; the global MATLAB path is not an Abaqus iteration algorithm.